compaction, sediment concentration in flowing streams, and to induce X-ray fluorescence in chemical analysis (§7.8). As an α-particle emitter, ^{241}Am has been mixed with beryllium to make intense neutron sources for oil well logging and for measuring water content in soils and in process streams in industrial plants. It is extensively used for elimination of static electricity and in smoke detectors where its use depends on the ionization of air.

^{252}Cf ($t_{½}$ 2.73 y) has found several uses. It decays to 3.1% by spontaneous fission (the main decay mode is through α emission) leading to a neutron emission rate of 2.3×10^{15} n s^{-1} kg^{-1} (average n-energy 2.35 MeV). The n-dose rate is 22 kSv h^{-1} kg^{-1}. ^{252}Cf is the only nuclide that can provide a useful neutron intensity over a sufficiently long half-life to make it a useful single nuclide neutron source. The low rates of heat emission (38.5 kW/kg), γ-radiation (initially 1.3×10^{16} photons s^{-1} kg^{-1}, yielding an initial dose rate of 1.6 kGy h^{-1} kg^{-1}; γ-ray intensity increases with time due to fission product build-up) and helium evolution (from α decay) allow fabrication of simple, small ^{252}Cf sources that require no external power supply nor any maintenance but that can provide moderately high neutron fluxes. The 2003 production rate of ^{252}Cf from the Oak Ridge facilities (HFIR-TRU) was ~ 250 mg/y and from russian Dimitrograd ~25 mg/y. Among the applications of ^{252}Cf we list the following:

(a) process control by a variety of on-stream non-destructive analytical techniques;
(b) medical diagnosis by activation analysis;
(c) production of short-lived radioisotopes at locations where they will be used, thus avoiding decay during transportation from an accelerator or reactor at another site;
(d) industrial neutron radiography, which images low density materials — especially hydrogenous materials — better than X-rays;
(e) possible medical treatment of tumors with ^{252}Cf sources that can be implanted in the body;
(f) petroleum and mineral exploration in which the compactness and portability of ^{252}Cf neutron sources facilitate testing for valuable deposits, particularly in inaccessible places such as deep wells and the sea floor;
(g) moisture measurements;
(h) hydrology studies to locate sources of water;
(i) nuclear safeguards tests, e.g. for criticality control in reactor fuel storage areas and for nuclear materials accountability (detection and recovery of fissionable material; implementation of nuclear agreements).

14.5. TRANSACTINIDE PROPERTIES

The transactinide elements (sometimes also called Super Heavy Elements) have only been produced a few atoms at a time, When the number of atoms observed in an experiment is very small the usual fundamental thermodynamic descriptions no longer

hold. Thus, the phase concept disappears when the medium contains only one or a few radioactive atoms. The law of mass action is also not applicable. In solvent extraction the distribution ratio between two phases loses its significance: a single atom can only exist in one of the two phases at the moment of separation. An average distribution can, however, be obtained by performing a large number of identical experiments. The number of decay events in each phase is recorded. The decay statistics are replaced by the statistical distribution of experimental results.

It is only possible to make compounds that contain one of the rare atoms. Thus, it is possible to make $Rf(NO_3)_4$ but it will probably never be possible to synthesize a compound like Rf_3N_4. Disproportionation is impossible since it involves exchange of electrons between two atoms of the same element. Thus, studies of the heaviest elements involve a lot of statistical considerations.

14.5.1. Chemistry of the transactinide elements

Studies of chemical properties of the transactinide elements are hampered by the short half-lives of most of the isotopes and the few atoms produced. The production cross sections are very small (0.01−0.5 nb) so chemical studies must be done very rapidly on one atom at a time. The detection systems must allow alpha-alpha correlation analysis. However, these elements offer a possibility to investigate the influence of relativistic effects of the electrons which could alter the relative stability of the 7s, 6d and 7p valence electrons. The result would be the existence of the elements in oxidation states other than those predicted from their expected position in the periodic table, see App. I.

Rutherfordium has been studied in the gas phase as well as in aqueous solution. The experiments confirm the tetravalent nature of Rf. Moreover Rf resembles Zr(IV) and Hf(IV) more closely than Th(IV), as expected for a transition element.

Chemical studies in aqueous solution and in the gas phase of dubnium indicated its similarity to the Group 5 transition elements niobium and tantalum, for which +5 is a stable oxidation state, hence a pentavalent nature of element 105 can be assumed. It resembles Nb(V) more than Ta(V). There are also some similarities between Db(V) and Pa(V), which are not expected from a simple extrapolation of Group 5 properties to $Z = 105$.

The limited aqueous chemistry data available on seaborgium imply that Sg has a stable hexavalent oxidation state and that Sg(VI) resembles Mo(VI) and W(VI). This is also indicated by gas phase chemistry data.

The bohrium compound BhO_3Cl has been synthesized in the gas phase using the isotope ^{267}Bh. This compound shows that Bh behaves like its lighter homolog Re, i.e. it belongs to group 7.

The first hassium compound to be synthesized in a gas phase experiment was HsO_4. This places Hs in group 8, below Ru and Os. To verify the group assignment sodium hassate, $Na_2HsO_4(OH)_2$ was successfully synthesized in another experiment.

No chemical studies have so far been performed on meitnerium, darmstadtium or roentgenium.

The chemical behaviour of copernicium has been studied in gas phase experiments where Cn has been allowed to adsorb on surfaces in a system with alpha detectors at different temperature. The temperature gradient goes from room temperature to the temperature of liquid nitrogen (−196 °C). The results imply that Cn is very volatile but it still has the adsorption characteristics of a metal. It is therefore assumed that it belongs to group 12.

No chemical studies have so far been performed on element 113.

The chemical behaviour of element 114 has been studied in gas phase experiments where 114 has been allowed to adsorb in a system with alpha detectors at different temperature. The temperature gradient goes from room temperature to the temperature of liquid nitrogen (−196 °C). The results imply that 114 is very volatile (even more volatile than Cn) but it still has the adsorption characteristics of a metal. It is therefore assumed that it belongs to group 14. Cn and 114 can both be characterized as "gaseous metals". Cn and 114 are the first elements where relativistic effects significantly alter the chemical properties.

No chemical studies have so far been performed on elements 115, 116, 117 and 118.

It should be emphasized that many of the results in this region are based on very few decay events, in some cases only two atoms, due to the low production rates and short half-lives. This gives large uncertainties and there is room for surprises in future experiments. The only trtansactinide elements where a significant amount of data has been collected with different methods in independent experiments are Rf and Db.

14.6. THE END OF THE PERIODIC TABLE

A question that arises in heavy element research is: how far can we reach? Is there a limit on the number of elements that cannot be exceeded? The answer is that theoretically it is possible to find quite a number of new elements above $Z = 118$ but in practice the number of elements will be limited by experimental difficulties. To reach far beyond $Z = 125$ and approach $N = 184$ will require even more neutron-rich actinide targets and more neutron-rich projectiles, e.g. short-lived fission products. The limit of hot fusion may thus be reached short of arriving at the proposed magic region around $N = 184$.

14.7. EXERCISES

14.1. What nuclear reactions would be suitable to make gold?

14.2. What fraction of neptunium is in the +4 state in a 0.1 M $Fe(SO_4)_2$ solution of acidity 1 M H^+, assuming that Np initially is present as Np(V) at a total concentration of 9.4 mM?